O11e - Optical Polarisation

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Abstract

The investigation into the intensity variation of polarized light as a function of analyser angle for three orientations of a $\lambda/4$ plate: $\varphi = 0°$, $30°$, and $45°$. Experimental measurements were analyzed and fitted against theoretical models derived from Jones Calculus. For $\varphi = 0°$ and $\varphi = 30°$, the models demonstrated strong quantitative agreement with experimental data, yielding high correlation coefficients ($r^2 = 0.9982$ and $r^2 = 0.9959$ respectively). Inclusion of amplitude, intercept, and phase shift as fitting parameters allowed for correction of minor experimental deviations, with amplitude values remaining near unity and intercept values effectively zero, indicating minimal systematic error. A significant phase shift was observed at $\varphi = 30°$, likely resulting from optical path length differences between theoretical assumptions and experimental configuration. For $\varphi = 45°$, the experimental data diverged from theoretical expectations of constant intensity, attributed to practical challenges in achieving perfect circular polarization, including alignment uncertainties and polarizer imperfections. Although analyser angle uncertainties were considered during measurements, they were not incorporated into the theoretical uncertainty propagation. In the second part, the polarization states of three unknown optical components were determined. Two of the elements (BBD2 and BBD3) were successfully classified through qualitative analysis of the intensity profiles: one transmitted linearly polarized light (BBD2), while the other (BBD3) produced elliptically polarized light. For the third component (BBD1), visual analysis was inconclusive due to weak intensity modulation. Therefore, a quantitative approach based on the Stokes parameters was employed. The resulting Stokes vector yielded a degree of polarization of approximately 16.9%, indicating that the light was primarily unpolarized with a weak elliptically polarized component. The experiment demonstrated both the effectiveness and limitations of qualitative polarimetry and highlighted the value of the Stokes formalism in resolving ambiguous polarization states. Together, these methods provided a comprehensive understanding of the behaviour and analysis of polarized light.

1 Introduction

2 Theoretical Analysis

(Task 0)

Jones Calculus provides a mathematical framework for analyzing polarized light transformations. It represents changes in amplitude and phase using Jones vectors for polarization states and Jones matrices for optical elements. For a wave propagating in the z-direction, the electric field is expressed as [1]:

$$\vec{J} = \frac{1}{|\vec{E}|}\begin{pmatrix} E_{0x}e^{-i\delta_x} \\ E_{0y}e^{-i\delta_y} \end{pmatrix}$$

where $\delta_x = 0$ by convention, making $\delta = \delta_y$ the physically relevant phase difference. The intensity is proportional to the squared magnitude of the output Jones vector.

2.1 Case 1: Linearly Polarized Light

2.1.1 Incident Light and Setup

Linearly polarized light with polarization angle α from vertical:

$$\vec{J}_\alpha = \begin{pmatrix} \sin\alpha \\ \cos\alpha \end{pmatrix}$$

Analyser matrix at angle θ from vertical [2]:

$$M_\theta = \begin{pmatrix} \sin^2\theta & \sin\theta\cos\theta \\ \sin\theta\cos\theta & \cos^2\theta \end{pmatrix}$$

2.1.2 Transmitted Intensity

Output electric field:

$$\vec{E}_{\text{out}} = M_\theta \vec{J}_\alpha = \cos(\alpha - \theta)\begin{pmatrix} \sin\theta \\ \cos\theta \end{pmatrix}$$

Intensity follows Malus' law:

$$I(\theta) \propto |\vec{E}_{\text{out}}|^2 = \cos^2(\alpha - \theta)$$

Maximum at $\theta = \alpha$, zero at $\theta = \alpha \pm 90°$.

2.2 Case 2: Circularly Polarized Light

2.2.1 Incident Light and Setup

Right-circular polarized light (clockwise rotation when facing source) [1] [2]:

$$\vec{J}_+ = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix}$$

Same analyser matrix M_θ.

2.2.2 Transmitted Intensity

Output electric field:

$$\vec{E}_{\text{out}} = M_\theta \vec{J}_+ = \frac{1}{\sqrt{2}}\begin{pmatrix} \sin\theta(\sin\theta - i\cos\theta) \\ \cos\theta(\sin\theta - i\cos\theta) \end{pmatrix}$$

Constant intensity:

$$I(\theta) \propto |\vec{E}_{\text{out}}|^2 = \frac{1}{2}$$

Independent of θ and φ.

2.3 Case 3: Linearly Polarized Light Through $\lambda/4$-Plate

2.3.1 Setup

Input: $\vec{J}_\alpha = \begin{pmatrix} \sin\alpha \\ \cos\alpha \end{pmatrix}$

$\lambda/4$-plate with fast axis at φ from vertical:

$$M^{\varphi}_{\lambda/4} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 + i\cos(2\varphi) & -i\sin(2\varphi) \\ -i\sin(2\varphi) & 1 - i\cos(2\varphi) \end{pmatrix}$$

2.3.2 Transmitted Intensity

After $\lambda/4$-plate:

$$\vec{E}_{\text{mid}} = M^{\varphi}_{\lambda/4}\vec{J}_\alpha$$

After analyser:

$$\vec{E}_{\text{out}} = M_\theta \vec{E}_{\text{mid}}, \quad I(\theta) \propto |\vec{E}_{\text{out}}|^2 \tag{1}$$

2.3.3 Special Cases

1. $\alpha = \varphi$: Remains linear at φ

$$I(\theta) \propto \cos^2(\varphi - \theta)$$

2. $\alpha = \varphi \pm \pi/2$: Linear at $\varphi \pm 90°$

$$I(\theta) \propto \sin^2(\varphi - \theta)$$

3. $\alpha = \varphi \pm \pi$: Equivalent to $\alpha = \varphi$

$$I(\theta) \propto \cos^2(\varphi - \theta)$$

4. $\alpha = \varphi \pm \pi/4$: Converts to circular

$$I(\theta) \propto \frac{1}{2} \quad \text{(constant)}$$

2.4 Case 4: Circularly Polarized Light Through $\lambda/4$-Plate

2.4.1 Setup

Right-circular input $\vec{J}_+$ through $\lambda/4$-plate at φ, then analyser at θ.

2.4.2 Transmitted Intensity

Converts to linear polarization at angle:

$$\beta = \varphi - \frac{\pi}{4} \quad \text{(from vertical)}$$

Intensity variation:

$$I(\theta) \propto \cos^2(\beta - \theta) = \cos^2\left(\varphi - \frac{\pi}{4} - \theta\right)$$

2.4.3 Special Cases

1. $\varphi = \pi/4$: Vertical polarization

$$I(\theta) \propto \cos^2\theta$$

2. $\varphi = -\pi/4$: Horizontal polarization

$$I(\theta) \propto \sin^2\theta$$

3 $\lambda/4$-Plate and Light Intensity

(Task I)

3.1 Experimental Setup

The experiment was designed to investigate the effect of a quarter-wave ($\lambda/4$) plate on the polarization state of linearly polarized light, as well as the resulting intensity modulation observed after an analyser. The optical setup consisted of a collimated LED light source emitting light with a peak wavelength of approximately 590 nm, followed by a linear polarizer that ensured the emitted light was linearly polarized in a fixed direction. This polarized light then passed through a rotatable $\lambda/4$ plate mounted on a rotation stage, followed by a second linear polarizer acting as an analyser. The analyser was also mounted on a rotatable stage, allowing its transmission axis to be varied through a full angular range. The light transmitted through the analyzer was focused onto a photodiode using a converging lens, and the resulting photocurrent was measured with a digital multimeter. Since the photodiode current is proportional to the incident light intensity, this allowed for quantitative measurements of intensity as a function of analyser and $\lambda/4$ plate orientation.

Before the measurements were taken, a calibration step was performed to define a consistent angular reference for both the analyser and the $\lambda/4$ plate. First, the $\lambda/4$ plate was removed from the beam path. The analyser was then rotated until the transmitted light intensity, as observed on the multimeter, reached a maximum. This indicated that the analyser's transmission axis was aligned with the polarization direction of the incoming light. The $\lambda/4$ plate was then reinserted and rotated such that its fast axis was also aligned with this direction. This orientation was subsequently defined as the zero angle ($\varphi = 0°$ for the $\lambda/4$ plate and $\theta = 0°$ for the analyser) for all further measurements. This alignment ensured that all angle values used during the experiment were defined relative to the initial polarization direction, thereby allowing for direct comparison with theoretical predictions without the need for angular corrections. All optical components were carefully aligned on an optical bench to minimize misalignment and background interference.

To capture the full angular dependence of the transmitted intensity, measurements were taken for each orientation of the quarter-wave plate at 20 distinct analyser angles. A step size of 10° was used across most of the range, while a finer step of 5° was employed near the edges of the mechanically accessible rotation range in order to include the final positions on the angular scale. This was repeated for $\varphi = 30°$ and $\varphi = 45°$.

The experimental data collected was then modeled to the relevant equations as detailed in the theoretical analysis for $\varphi = 0°$, $\varphi = 30°$ and $\varphi = 45°$, outlined below as:

$$\begin{aligned} \varphi = 0° : &\quad I(\theta) = A\cos^2(\theta) \\ \varphi = 30° : &\quad I(\theta) = \tfrac{1}{2} + \tfrac{1}{8}\cos(2\theta) + \tfrac{\sqrt{3}}{8}\sin(2\theta) \\ \varphi = 45° : &\quad I(\theta) = A\tfrac{1}{2} \end{aligned} \tag{2}$$

Where the solution for $\varphi = 30°$ was derived from Eq.(1).

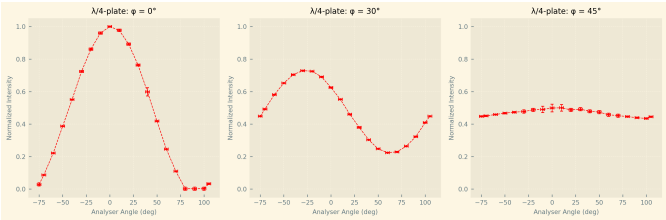

Figure 1: Enlarged Raw data of Intensity as a function of the analyser angle.

3.2 Data Analysis

The results of Fig. 5 show that as the angle φ tends to $\pi/4$ the amplitude and period of the intensity as a function of α decreases. This coincides with the theoretically expected results for Case 3. However for the case of $\varphi = \pi/4$ it is expected that the light wave is perfectly circular and thus I(θ) is constant. This is not seen in the experimental data, there is a definitive peak around 10°, and a trough at approximately 80°. To quantitatively assess the data, each case experimentally measured was modeled to its appropriate mathematical expression as outlined in the theoretical analysis.

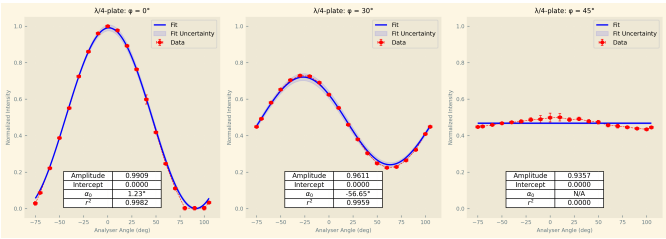

Figure 2: Fitted data of Intensity as a function of the analyser angle, with appropiate models as outlined in Eq.(2)

The quantitative analysis found a strong correlation, ($r^2 = 0.9992$, $r^2 = 0.9659$), for the $\varphi = 0°$ and $\varphi = 30°$ respectively, for the $\varphi = 45°$ fitting, the non-linear fitting results in a poor correlation between experimental and theoretical data. As the theoretical model is constant and does not depend on the analyser angle, α, the residuals seen in the experimental data starkly contrast with a constant linear line, and results in a quantitative result of $-inf$ which was defaulted to 0 to indicate no correlation. The experimental variation from theoretical values shown in the $\varphi = 45°$ case is mostly attributed to light bleeding through the two-polarizers as a result of factors such as uncertainty in alignment, manufacturing error in the polarizers construction and other practical complications encountered when building a polarizer.

It is also of note that normalized amplitude for modeled fitting was consistently close to 1 which reassures that the theoretical fitting aligns with experimental findings, the choice to add Amplitude as a variable parameter for the fitting was made to account for unexpected deviations between experimental results and theoretical analysis, that may arise from experimental oversight or error. It should also be considered that as there does still exist slight deviations in fitted amplitude across the φ angles, the determined r^2-values are expected to be slightly off from true-values.

The intercept was also included as a variable parameter from the same precautionary reasons as the amplitude, as its determined value across all datasets was a definitive 0.0000 it further reinforces that the there is no experimental oversight or error in the experimental measurements.

α_0 was included as it was expected that the experimental data would not be in-phase with theoretical analysis, This is because the wave-forms seen in Fig. 5, Fig. 6 and Fig. 7 represent intensity points of a spiraling, and oscillating electromagnetic wave of which its components vary with distant travelled and there is high probability that the distance the EM-Wave travels in the experiment will differ from that of the assumption made in the theoretical analysis. The fitting found that the $\varphi = 0°$ was close to in-phase with theoretical predictions with $\alpha_0 = 1.23°$, where as the $\varphi = 30°$ was significantly out-of-phase,$\alpha_0 = -56.65°$, and α_0 was not applicable for $\varphi = 45°$ as theoretical predictions were constant.

4 Determination of unknown polarizers
(Task II)

4.1 Experimental Setup

The experimental configuration for the polarization analysis of unknown optical elements (referred to as "black boxes") remained identical to the setup previously described, consisting of a collimated LED source, fixed linear polarizer, optional $\lambda/4$ plate, rotatable analyser, photodiode, and digital multimeter. However, in this case, the black boxes were inserted in the holder after the LED source instead of the linear polarizor, and their effect on the polarization state of light was investigated.

For each black box, two separate measurement sets were recorded: one without the quarter-wave plate and one with it. Prior to both sets of measurements, a calibration step was again performed by temporarily removing the $\lambda/4$ plate and rotating the analyser to the angle of maximum transmitted intensity. This angle, corresponding to the dominant output polarization direction after the black box, was then used as the fixed reference orientation for the $\lambda/4$ plate in the second measurement series, where it was reinserted and aligned accordingly. This ensured consistent and meaningful interpretation of the analyser scans with and without the $\lambda/4$ plate.

By comparing the measured intensity patterns across analyser angles in both configurations with the key identification properties provided in [1], the polarization state induced by each black box was characterised, distinguishing between unpolarized, linear, circular, elliptical, and partially polarized light.

4.2 Data Analysis

The following excerpt is taken directly from the lab manual [1]:

"If one has to analyze the unknown polarization state of a light beam, one might measure the four intensities defined above and calculate the Stokes parameters. On the other hand, the following simple guide might be of

help. This uses a rotatable linear analyser and a $\lambda/4$ – plate. If the intensity variation is monitored when the linear analyser is rotated, with the $\lambda/4$ – plate inserted before the analyser or not, the following conclusions can be drawn:

1. **No intensity variation with analyser alone** → Insert $\lambda/4$ – plate in front of analyser with fast axis in arbitrary direction →
   (a) No intensity variation for all analyser angles → *natural unpolarized light*
   (b) Intensity minimum for one analyser position →
      i. Zero intensity for one analyser position → *circularly polarized light*
      ii. Non-zero intensity for all analyser angles → *mixture of circularly polarized and unpolarized light*
2. **Intensity variation with analyser alone** →
   (a) Zero intensity for one analyzer position → *linearly polarized light*
   (b) Non-zero intensity for all analyser angles → Insert $\lambda/4$ – plate in front of analyzer with fast axis parallel to analyzer direction at maximum or minimum intensity →
      i. Zero intensity for one analyzer position → *elliptically polarized light*
      ii. Non-zero intensity for all analyzer angles →
         A. Minimum intensity at the same analyzer setting as without $\lambda/4$ – plate → *mixture of linearly polarized and unpolarized light*
         B. Minimum intensity at different analyzer setting as without $\lambda/4$ – plate → *mixture of elliptically polarized and unpolarized light.*"

Using these instructions, polarization states for each of the three black boxes were identified. Fig. 7 shows the plotted data for each of the black boxes (labelled: BBD1, BBD2 and BBD3) with and without the $\lambda/4$ – plate.

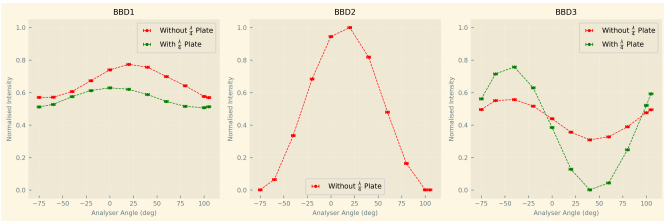

Figure 3: Enlarged Black Box Data for a range of analyser angles with and without the $\lambda/4$-plate

For BBD1, data shows that the first identification quality is certainly not true. The sequence of observed identification quality are as follows: 2 → (b) → ii → B. The last statement (B) is true since the minima for both measurement sets (with and without the $\lambda/4$ – plate) occure at approximately the same angle. Hence BBD1 transmits a mixture of elliptically polarized and unpolarized light.

Similar analysis was carried out for BBD2. No measurements were taken with the $\lambda/4$ – plate in this case since the first measurement set provided sufficient evidence to identify the polarization state of the black box. The sequence of identification was as follows: 2 → (a). This proves experimentally that BBD2 transmits linearly polarized light.

In the case of BBD3, analysis shows that the sequence is: 2 → (b) → (i), proving that BBD3 transmits elliptically polarized light.

4.2.1 Analysis of BBD1 Using the Stokes Parameter Method

For the analysis of the three unknown black boxes, the qualitative procedure outlined in [1] (based on intensity variation with and without a quarter-wave plate) proved sufficient in two cases (BBD2 and BBD3). In both of those, the intensity patterns observed allowed for an unambiguous classification (clear zero intensity points). However, for BBD1, the results of this qualitative method were inconclusive. Although, the observed minima seem to occur at the same angles for the case of angle variation in analyser only and for the case of the addition of the $\lambda/4$ - plate as well, it can not be conclusively stated that the aforementioned result is true conclusively. To gain a clearer understanding of the polarization state and to increase confidence in the previously obtained result, the Stokes parameter method was therefore applied for BBD1.

The Stokes parameters provide a complete and quantitative description of the polarization state of light. This is achieved through a set of four real-valued parameters:

- s_0: total intensity of the light,
- s_1: difference in intensity between horizontal and vertical polarization,
- s_2: difference between linear polarizations at $+45°$ and $-45°$,
- s_3: difference between right- and left-circular polarization components.

Together, they define the state of polarization and can also be used to calculate the degree of polarization (DOP), which distinguishes between fully, partially, and unpolarized light. The DOP is defined as:

$$\text{DOP} = \frac{\sqrt{s_1^2 + s_2^2 + s_3^2}}{s_0}$$

To obtain the Stokes parameters for BBD1, a series of intensity measurements were taken at fixed analyser angles. In hindsight, for some required angles (e.g., $\pm 45°$ and $90°$) due , direct values were not available, so nearby measurements (e.g., $40°, -40°, 80°, 100°$) were linearly interpolated.
The resulting Stokes vector was:

$$\vec{s} = \begin{pmatrix} s_0 \\ s_1 \\ s_2 \\ s_3 \end{pmatrix} = \begin{pmatrix} 0.569 \\ -0.055 \\ 0.061 \\ 0.050 \end{pmatrix}$$

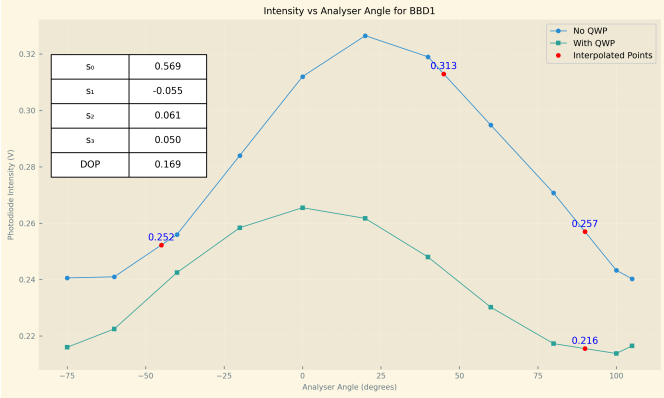

Figure 4: Application of Stokes Parameter Method on BBD1 for the Determination of Polarization State

This corresponds to a degree of polarization of approximately 16.9%, which is significantly below unity. This low DOP indicates that the light emerging from BBD1 is predominantly unpolarized, with only a small polarized component. The nonzero values of s_1, s_2, and s_3 show that the polarized portion is elliptically polarized, with a slight right-handed bias ($s_3 > 0$) and orientation near $+45°$ from the vertical axis ($s_2 > 0$).

In summary, the application of the Stokes parameter method revealed that BBD1 transmits a mixture of elliptically polarized and predominantly unpolarized light, a conclusion that agrees with the previously determined result.

5 Discussion

5.1 $\lambda/4$ - Plate and Light Intensity

Across the three different angles for the $\lambda/4$ plate, $\varphi = 0°, 30°, 45°$, the results strongly aligned with theoretical models and predication with $\varphi = 0°, 30°$ possessing correlations of $r^2 = 0.9982$ and $r^2 = 0.9959$ respectively. For the $\lambda/4$ plate at an angle of $\varphi = 45°$ the results did not quantitatively correlate to theoretical results, however after a qualitative investigation it was apparent that this is due to the linearly constant theoretical model, against a varying and dependent experimental finding. The difference between the two is believed to be a result of the practical difficulty in creating a perfect polarizer, along with a experimental misalignment/uncertainty in the analysers angular position, resulting in light bleeding through the apparatus and thus deviation significantly from a linearly constant expected result.

The inclusion of Amplitude and Intercept as variable parameters in the fitting process, despite using a normalized intensity, was a precautionary measure to account for experimental oversight or errors during measurements, however across all angles of φ the Intercept was determined to be 0.0000 and Amplitude remained close to 1, however the small deviations from 1 seen in the each dataset correlate to a residual-deviation that the respective r^2 does not reflect, adding a small-error to the determined r^2.

Optical-path-length distance differences between theoretical results and the experimental measurements required the introduction of a phase-shift to sensibly assess the validity of theoretical model with experimental results. The associated phase shifts for $\varphi = 0°$,$\varphi = 30°$ were determined to $\alpha_0 = 1.23°$ and $\alpha_0 = -56.65°$, a phase shift was not applicable to $\varphi = 45°$ as the theoretical model was constant.

Reviewing the error propagation used in this investigation of Intensity as a function of analyser angle for varying $\lambda/4$-plate angles all error considered was experimental or systematic error in measurements applied to the data. There was no attempt to incorporate the $\pm 2°$ uncertainty of the analyser angle into the models uncertainty and thus take into account the variable nature of the Jones Calculus model used. A model that varies drastically if either φ or θ vary. For future investigation it is recommended to incorporate the theoretical uncertainty as a result of the experimental uncertainty in measurements, to more accurately convey the uncertainty expressed in the experiment. Furthermore by removing the variable parameters Amplitude and Intercept, the expressed uncertainty will also more accurately convey the true uncertainty and thus confidence in the measurements determined.

Later, the polarization states of three unknown optical elements were examined by seeing how they affected the intensity of transmitted light through a rotatable linear analyser, both with and without a quarter-wave plate inserted. The identification process relied initially on observing characteristic intensity variations as described in the experimental guide, which outlines how specific modulation patterns correspond to linearly, circularly, or elliptically polarized light, as well as unpolarized or partially polarized mixtures.

For two of the optical elements, the classification was unambiguous. BBD2 produced a clear intensity minima, consistent with linearly polarized light. The observed cosine-squared dependence, with complete extinction at orthogonal angles, confirmed this without the need for additional analysis. BBD3 exhibited no complete extinction with only the analyser in place but showed a smooth variation in intensity with a point with almost zero intensity with quarter-wave plate inserted. This indicated the presence of elliptically polarized light, according to the guide provided in [1].

In contrast, BBD1 presented a more ambiguous case. The analyser intensity showed only slight modulation and no extinction, even with the quarter-wave plate inserted. This weak response made a definitive classification based on the qualitative pattern ,and the determination of the position of the minima for each case, (with and without the $\lambda/4$ - plate) alone unreliable. To provide an additional source of result to increase the confidence in previously determined polarization state, the Stokes parameter formalism was employed. This method allows for a full quantitative characterization of the polarization state by combining intensity measurements at several analyser angles into a four-component Stokes vector.

Using interpolated values for analyser positions not

directly measured, the Stokes vector for BBD1 was determined. The result revealed a degree of polarization of approximately 16.9%, which is relatively low. This confirmed that the light was primarily unpolarized, with only a small contribution from an elliptically polarized component, which agreed with the previous result. The signs and relative magnitudes of the Stokes parameters indicated a slight right-handed ellipticity oriented near $+45°$ from vertical.

This case highlighted the limitations of relying solely on visual criteria when dealing with partially polarized light or weak polarization signatures. While the qualitative guide is effective in clearly polarized cases, it lacks the resolution needed for mixed or borderline states. The Stokes method, though more demanding in terms of measurements, offers a mathematically rigorous and objective classification tool.

In summary, all three unknown optical elements were successfully characterized using a combination of qualitative and quantitative approaches. The experiment demonstrated that:

- BBD1 transmitted a mixture of elliptically polarized and largely unpolarized light,
- BBD2 transmitted linearly polarized light,
- BBD3 produced elliptically polarized light.

This layered analysis illustrates the importance of using both heuristic and formal tools in experimental polarimetry.

6 Conclusion

This experiment provided a comprehensive investigation into the behaviour of polarized light as it passes through various optical components. By combining theoretical predictions with experimental measurements, the experiment successfully demonstrated the principles of polarization, the function of a quarter-wave plate, and how different optical elements alter the polarization state of light.

The results of the investigation into the intensity as a function of analyser angle across the three $\lambda/4$-plate angles $\varphi = 0°, 30°, 45°$, found strong agreement with theoretical models for $\varphi = 0°$ and $30°$. For both of these cases the fitted models high correlated with experimental data $r^2 = 0.9982$ and $r^2 = 0.9959$ respectively, validating the theoretical expectations, dervied through the use of Jones Calculus and Maus' Law. The phase shift fitting parameters effectively accounted for differences in optical-path-length between theoretical assumptions and experimental execution. the addition of amplitude and intercept were intially chosen as fitting parameters to account for unexpected experimental errors, however proved to further reinforce the theoretical models, with small deviations seen in amplitude.
For $\varphi = 45°$, the theoretical model predicted a constant intensity across all analyser angles, which corresponds to a perfectly circular polarization, the experimental data however, showed a clear angular dependence. The deviation is attributed to practical limitations and difficulties in constructing a perfect polarizer, errors in aligning the analyser angle at exactly $45°$ and other experimental uncertainties which led to light bleeding through the apparatus. Overall the first part of the experiment successfully demonstrated the expected dependence of light intensity on analyser angle across multiple $\lambda/4$-plate orientations, within experimental error and practical limitations.

The second part of the experiment focused on determining the polarization state of light emerging from three unknown optical elements. In two of the cases (BBD2 and BBD3), qualitative analysis based on the angular intensity profiles—with and without a quarter-wave plate—was sufficient to identify the polarization states as linear and elliptical, respectively. However, in one case (BBD1), the analysis was inconclusive due to minimal intensity modulation and ambiguity in the location of minima. To address this, the Stokes parameter method was applied. This quantitative technique revealed a low degree of polarization (approximately 16.9%), indicating that the light was predominantly unpolarized with a minor elliptically polarized component. This result confirmed and refined the initial qualitative classification.

Overall, the experiment demonstrated not only the theoretical principles behind polarization optics but also highlighted the importance of combining qualitative reasoning with quantitative methods when interpreting experimental data. The layered approach used—consisting of visual pattern recognition and Stokes vector formalism—proved essential in resolving ambiguous cases and ensured a robust characterisation of each optical system.

References

[1] Fakultät für Physik und Geowissenschaften, Physikalisches Grundpraktikum. *O11e Optical Polarisation*, June 2024. Laboratory manual.

[2] Jordan Louis Edmunds. Jones matrices. YouTube playlist, 2020. https://www.youtube.com/playlist?list=PLQms29D1RqeIJUzgzFweiuVFWPOtSNCUP, accessed June 22, 2025.

Appendix 1. Enlarged Figures Used in Report

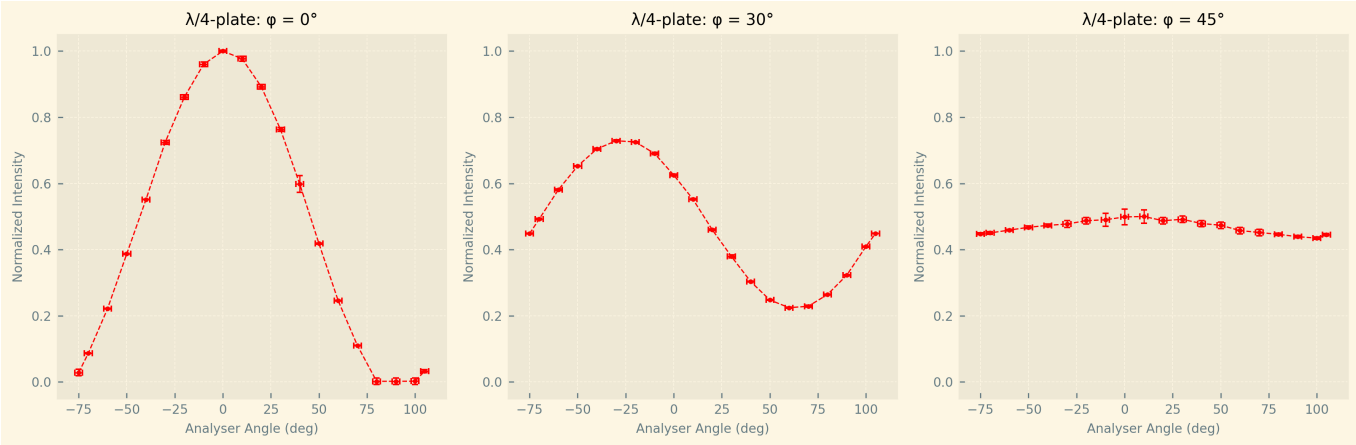

Figure 5: Enlarged Raw data of Intensity as a function of the analyser angle.

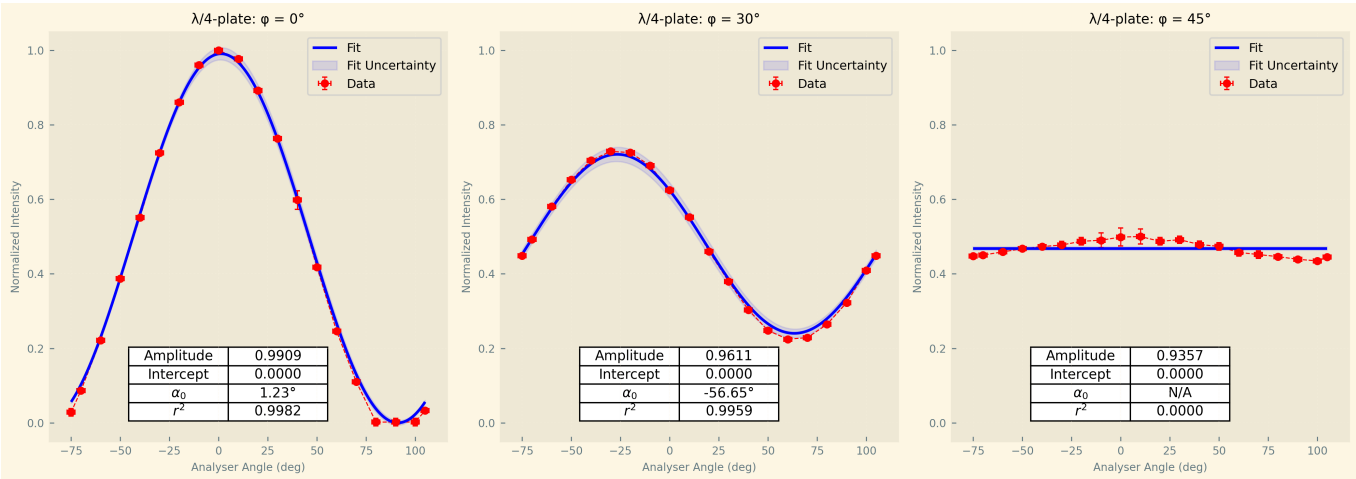

Figure 6: Enlarged Fitted data of Intensity as a function of the analyser angle, with appropiate models as outlined in Eq.(2)

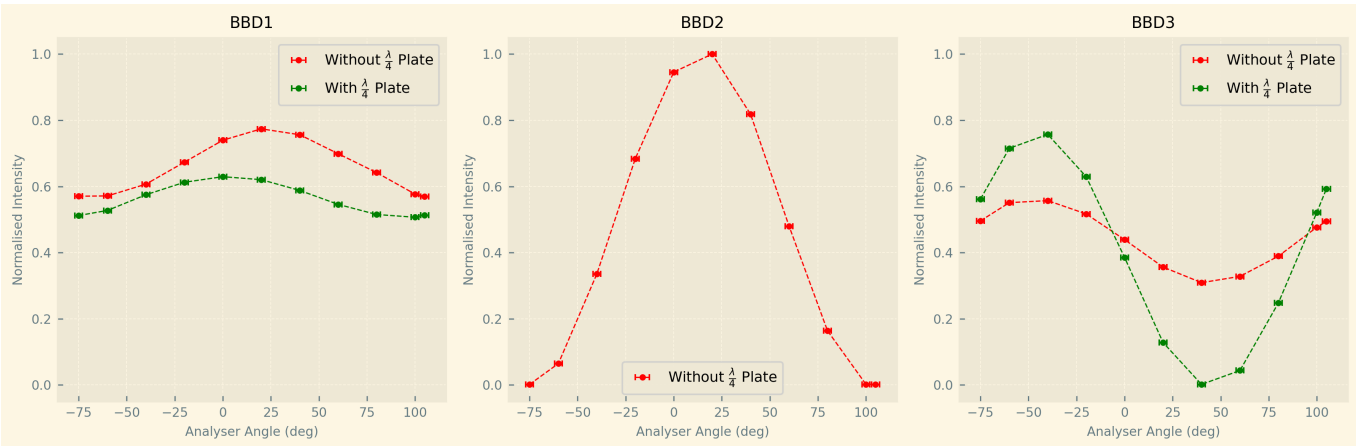

Figure 7: Enlarged Black Box Data for a range of analyser angles with and without the $\lambda/4$-plate

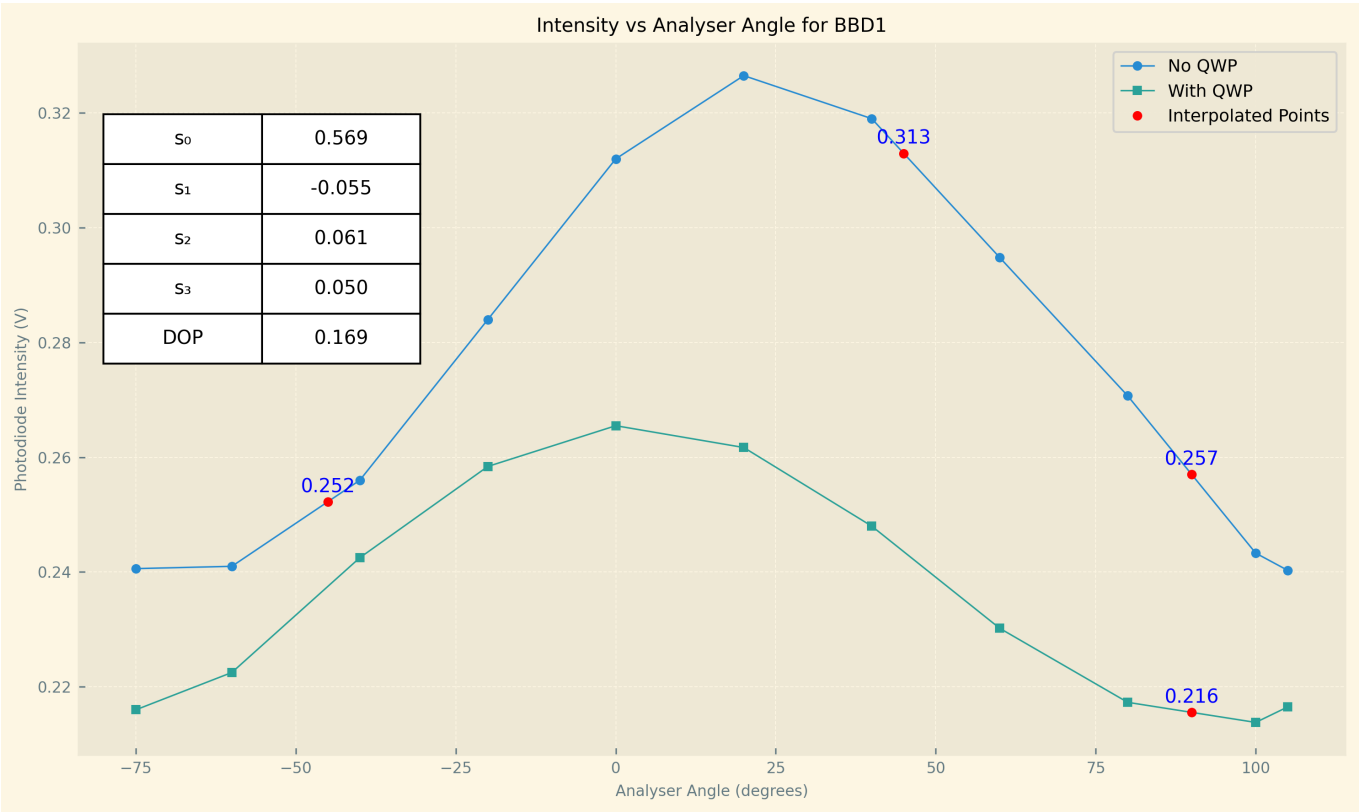

Figure 8: Application of Stokes Parameter Method on BBD1 for the Determination of Polarization State